

Energy Stock Price Forecasting

Using LSTM Neural Networks

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Stock	Data	Models	Framework
NTPC.NS (Nat'l Thermal Power)	5 years daily OHLCV data	Linear Reg, Random Forest, LSTM	PyTorch + scikit-learn

1. Project Overview

This project applies deep learning to forecast next-day closing prices of **NTPC** — India's largest power utility and the highest-weight constituent of the NIFTY Energy Index. The pipeline spans raw data collection through feature engineering, model training, and backtesting, demonstrating practical quantitative finance and machine learning skills.

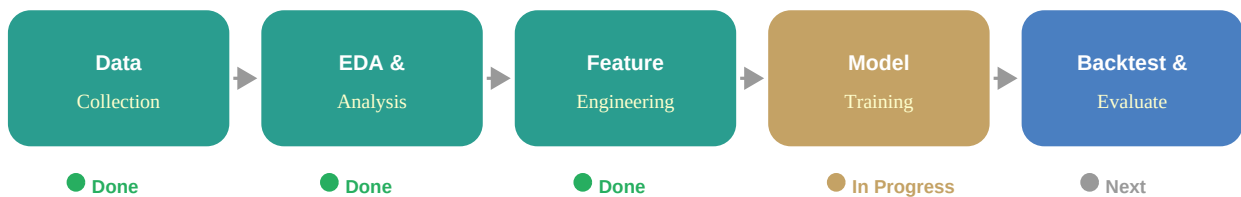
1.1 Why NTPC?

NTPC is the best single-stock proxy for India's energy sector. Its high liquidity and institutional coverage ensure clean, gap-free daily data over the 5-year study period.

1.2 Why LSTM?

Stock prices exhibit temporal dependencies — today's price depends on patterns in preceding days. LSTMs capture these sequential relationships through their gating mechanism (forget, input, output gates), selectively retaining or discarding information across time steps. Traditional models like Linear Regression and Random Forest treat each day independently and miss these dynamics.

2. Pipeline



3. Key Design Decisions

Decision	Choice	Rationale
Data split	70 / 15 / 15 chronological	No shuffling — prevents future data leakage
Feature scaling	StandardScaler on train only	Val/test must not influence normalization
Target scaling	Separate StandardScaler	LSTM needs input/output on same scale
Lookback window	30 trading days (~6 weeks)	Enough context for pattern recognition
Architecture	2-layer LSTM, 64 hidden units	Balances model capacity vs. overfitting risk
Early stopping	Patience = 10 epochs	Stops training when val loss plateaus
Gradient clipping	max_norm = 1.0	Prevents exploding gradients in LSTM

6. LSTM Architecture

Component	Configuration	Purpose
Input layer	17 features × 30 timesteps	30-day sliding window of technical indicators
LSTM Layer 1	64 hidden units	Learns short-term temporal patterns
LSTM Layer 2	64 hidden units, dropout = 0.2	Captures higher-order sequence dependencies
Dense layer	64 → 32 neurons, ReLU activation	Non-linear feature compression
Dropout	Rate = 0.1	Regularization before output
Output layer	32 → 1 (linear)	Predicts next-day closing price (scaled)
Optimizer	Adam, initial LR = 0.001	Adaptive per-parameter learning rate
LR Scheduler	ReduceLROnPlateau, patience = 5	Halves LR when validation loss stalls
Gradient clipping	max_norm = 1.0	Prevents exploding gradients in recurrent layers
Early stopping	Patience = 10 epochs	Saves best model, stops when overfit begins

7. Project Timeline — 3 Weeks (Aug 1 – 21)

Week	Period	Tasks	Status
1	Aug 1–2	Project setup, data download, EDA (6 plots), feature engineering, baseline models, LSTM training	Done
1	Aug 3–4	Backtesting (Sharpe ratio, max drawdown), debug LSTM, GitHub push	Next
1	Aug 5–7	Code review, add comments, understand model tradeoffs	Todo
2	Aug 8–9	Write README.md with results and embedded plots	Todo
2	Aug 10–11	Tune LSTM hyperparameters, try more stocks (Reliance, ONGC)	Todo
2	Aug 12–14	Study LSTM internals (gates, cell state), Sharpe derivation	Todo
3	Aug 15–17	Optional: crude oil prices as external feature, GRU comparison	Todo
3	Aug 18–19	Final code cleanup, pin repo on GitHub, update resume	Todo
3	Aug 20–21	Mock interview: explain project end-to-end in 5 minutes	Todo

8. Technical Skills

Domain	Tools & Concepts
Data Engineering	yfinance API, pandas, NumPy, pathlib, CSV data pipelines

Domain	Tools & Concepts
Machine Learning	scikit-learn (Linear Regression, Random Forest), StandardScaler, chronological split
Deep Learning	PyTorch, LSTM, DataLoader, early stopping, LR scheduling, gradient clipping
Visualization	matplotlib, seaborn (histplot, heatmap, KDE), statsmodels (ACF/PACF)
Statistics	ADF stationarity test, skewness, kurtosis, rolling volatility, max drawdown
Quant Finance	Sharpe ratio, backtesting, long-only strategy, directional accuracy, RMSE/MAE
Version Control	Git, GitHub, .gitignore, virtual environments, requirements.txt

GitHub: github.com/aariiparekh3012-collab/project1-aarya